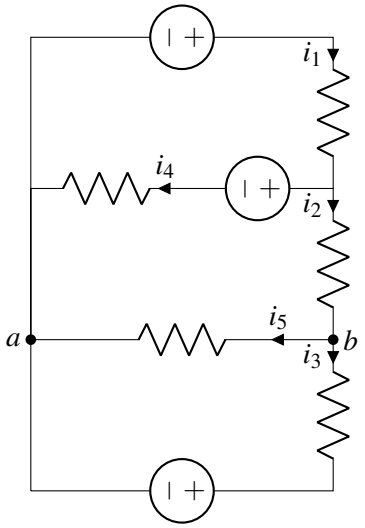
CSM
Fall 2025

EECS 16B
Worksheet 2

1. Resistive Circuit

Consider the following circuit network:



Where each voltage source produces a voltage V, and each resistor has a resistance R.

(a) Label the current through and voltage across each circuit component using the passive sign convention.

(b) Write down five linearly independent equations using KCL and KVL.

(c) Solve for the voltage difference ΔV_{ab} between points a and b.

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2. Thevenin's and Norton's Equivalent Circuit

In this set, we will review two useful theorems that you might have learned in previous physics classes. **Thevenin's Theorem** states that we can replace entire network by an equivalent circuit that contains only an independent voltage source in series with an impedance (resistor) such that the current-voltage relationship at the load is unchanged.

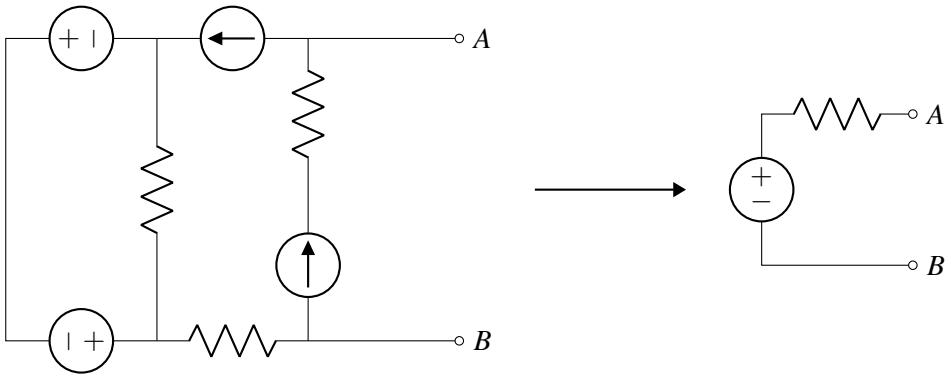


Figure 1: Thevenin's Equivalent Circuit

As shown above, we have a linear electrical network, containing only voltage sources, curretn sources, resistors, and two terminals A and B. The circuit can be always transformed to an equivalent circuit with only one voltage source, one resistor, and two terminals A and B. By using this theorem, we can largely simplfy the circuit. A more easier way to understand this is that we can use a black box to include everything connecting to two terminals, and then replace the black box with a voltage source and a resistor in series.

Norton's Thereom is identical to Thevenin's Theorem except that the equivalent circuit is an independent current source in parallel with an impedance (resistor). Therefore, the Norton equivalent circuit is a source transformation of the Thevenin equivalent circuit.

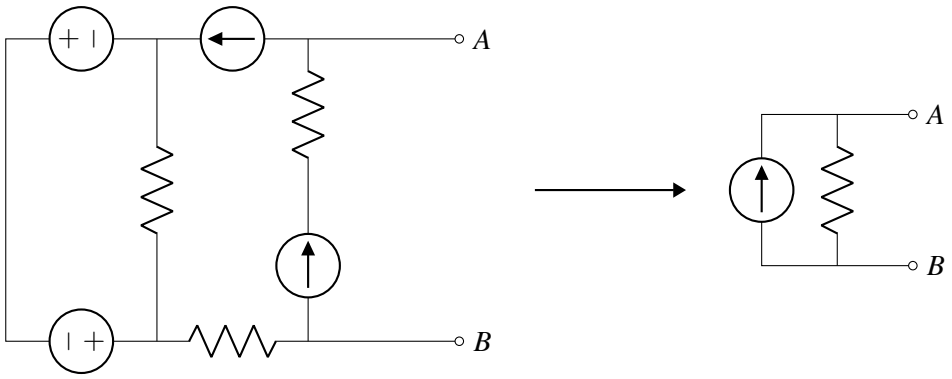


Figure 2: Norton's Equivalent Circuit

Quite similar to Thevenin's theorem, we can replace a linear electrical network with a current source and a resistor in parallel.Note that, the power source can be non-constant, which means that we could have an AC circuit. As a result, we need to use impedances at a given frequency substituting the resistances.

First, let's assume that the voltage source is ideal and does not have any internal resistance. In a non-ideal case, we need to add one more resistor beside the voltage source to represent the internal resistance of the voltage source. However, for an ideal current source, it has infinite internal resistance. Then use KVL, KCL,

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and Ohm's Law to find out the resistance and voltage in the Thevenin's equivalent circuit. Similar procedure also applies to the Norton's equivalent circuit.

For the following problems, we will first try to obtain the Thevenin's equivalent circuit of the following circuit.

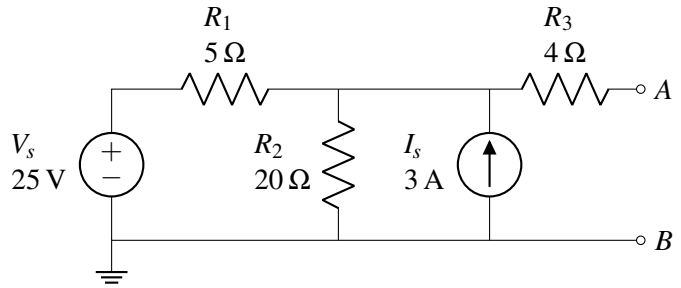


Figure 3: Circuit to examine

(a) First, we need to create an imaginary wire connecting to terminal A and B, and also a terminal C.

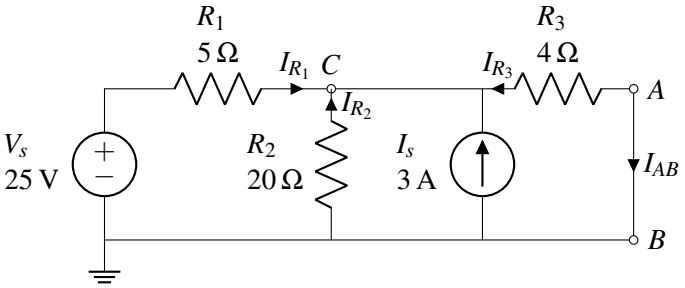


Figure 4: Determining short-circuit current

Denote the voltage at terminal C as V_c. What are the magnitudes of the currents I_{R_1}, I_{R_2}, I_{R_3}? By using KCL, can you derive an equation including all four currents?

(b) After establishing the equation, you should be able to find out V_C. Based on the value of V_C, what is the magnitude of I_{ab}? I_{ab} is called **short-circuit current**, it is the current that actually flow from A to B if you connect two terminals in the equivalent circuit.

(c) Then, suppose that there is **no such imaginary wire across A and B**. Let's focus on finding the value of the **Thevenin Equivalent Resistance** R_{th}. To calculate the Thevenin equivalent resistance, remove all power sources from the original circuit. The voltage sources are short-circuited and current sources are opened. What is R_{th}, the total resistance between terminal A and B in the remaining circuit?
Hint: The remaining circuit now only has three resistors. R_1 and R_2 are now in parallel and R_3 is in series with R_1 and R_2.

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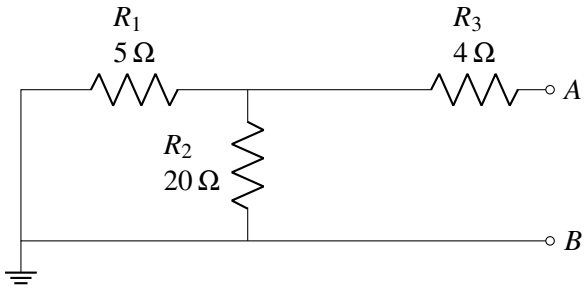


Figure 5: Circuit with nulled sources

(d) We are almost done! Now, it's time to draw our Thevenin's equivalent circuit.

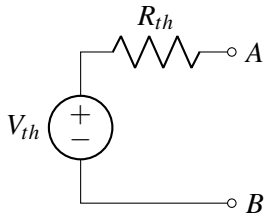


Figure 6: Thevenin equivalent circuit for this problem

What are the values of V_{th} and R_{th}?
Hint: What law should we use to get V_{th}? Remember I_{ab} is the short-circuit current.

Now we know how to get a Thevenin's Equivalent Circuit, but how do we get a Norton's Equivalent Circuit? In this question we will look at how to use Thevenin's Circuit to get the Norton's Circuit.

(e) Let's look at the same graph, Figure 3. You have already obtained the value of the short-circuit current I_{ab} and the total resistance R_{th} between the terminal A and B. Why don't we just use both of the values to find out I_{th} and R_{th} in the Norton's equivalent circuit shown below?

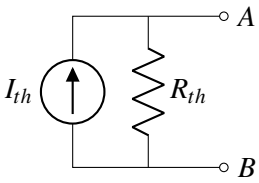


Figure 7: Norton equivalent circuit for this problem

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3. Complex Number Arithmatic

(a) For two complex numbers in Cartesian Form, $z_1 = a + jb$ and $z_2 = c + jd$:

i. What is $|z_1|$?
ii. What is $\angle z_1$?
iii. What is $z_1 + z_2$?
iv. What is $z_1 - z_2$?
v. What is $z_1 \cdot z_2$?
vi. What is z_1 / z_2?

(b) For two complex numbers in Polar Form, $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$:

i. What is $|z_1|$?
ii. What is $\angle z_1$?
iii. What is $z_1 + z_2$?
iv. What is $z_1 - z_2$?
v. What is $z_1 \cdot z_2$?
vi. What is z_1 / z_2?

(c) For two general complex numbers z_1 and z_2, answer in terms of $|z_1|$, $|z_2|$, $\angle z_1$, and $\angle z_2$:

i. What is $|z_1 \cdot z_2|$?
ii. What is $|z_1 / z_2|$?
iii. What is $\angle(z_1 \cdot z_2)$?
iv. What is $\angle(z_1 / z_2)$?

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